Closed-Loop and Activity-Guided Optogenetic Control

Paper: Grosenick, Marshel and Deisseroth (2015). Review of feedback, sensing and optical actuation.

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Page references below use PDF page numbers, including front matter.

To which general question does this paper relate?

What can be measured and controlled optogenetically, and how to formulate an experiment-specific control objective and model.

How does it answer the general questions?

Framework. Distinguishes feedback based on ongoing measurements from stimulation patterns chosen using earlier observations. Reviews electrical recordings, optical indicators and cell/projection-targeted light actuation, alongside on/off, PID, adaptive and predictive control approaches (PDF pp. 1–12).

Usable insight. The desired neural or behavioural trajectory determines sensor bandwidth, actuator choice and the meaning of successful control. Calcium signals do not necessarily resolve individual spike timing; opsin kinetics, saturation, targeting, heating and nonstationarity constrain achievable responses. Optogenetic actuation does not itself provide a neural measurement.

Model relevance. Supports system identification and explicit sensor/opsin dynamics before feedback design. A memoryless input map or time-invariant delayed rate model is a chosen approximation, not a property guaranteed by the technology. Examples include seizure interruption and manipulation of oscillations, but no universal beta-suppression objective or PIE Zames–Falb optimization is provided (PDF pp. 10–12, 24–26).

What papers does it cite that could provide more information?

- Paz et al. (2013), *Closed-loop optogenetic control of thalamus as a tool for interrupting seizures after cortical injury* (references, PDF p. 32).
- Krook-Magnuson et al. (2014), *Cerebellar directed optogenetic intervention inhibits spontaneous hippocampal seizures in a mouse model of temporal lobe epilepsy* (PDF p. 30).